

Lithium-ion battery charging management considering economic costs of electrical energy loss and battery degradation



Kailong Liu^a, Xiaosong Hu^{b,*}, Zhile Yang^{c,*}, Yi Xie^b, Shengzhong Feng^c

^a WMG, The University of Warwick, Coventry CV4 7AL, United Kingdom

^b Department of Automotive Engineering, Chongqing University, Chongqing 400044, China

^c Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong 518055, China

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ABSTRACT

Technical challenges facing the development of battery economic charging for energy management arise from various contradictory objectives, immeasurable internal states, and hard constraints. Available solutions often resort to optimizing economic charging from the perspective of power grid, while few focus on the benefits of electric vehicle (EV) owners. This paper proposes a constrained multi-objective optimization framework to achieve economy-conscious battery charging management. Specifically, a coupled electrothermal-aging model is first applied to capture the nonlinear electrical, thermal, and aging dynamics of a lithium-ion battery with different timescales. Then an economic indicator considering the total charging cost caused by both the battery aging and electrical energy loss is formulated, based on a battery resale cost model and the Beijing peak-valley time-of-use (TOU) price. In addition, two crucial but contradictory objectives including the charging time and battery average temperature are also taken into account, along with hard constraints on the battery State of Charge (SOC), charging current, terminal voltage, and temperature. The complicated multi-objective charging problem is solved by using NSGA-II to optimize the multistage constant-current (MCC) profile. Sensitivities of the cut-off voltage, convection resistance, and ambient temperature are comparatively analysed via Pareto frontier. Numerical results of two extreme cases and several balanced charging cases demonstrate that feasible, economy-conscious battery charging management for various user demands can be achieved through suitable equilibrations among charging speed and thermal variations during charging.

1. Introduction

Electric vehicles (EVs) have been being developed rapidly owing to their enormous promises in energy saving and environmental benefits. Lithium-ion (Li-ion) batteries are major energy storage sources supplying power for EVs due to their outstanding features such as high energy density, no memory effect, and environmental friendliness [1,2]. However, several problems remain to be further solved for enhancing the performance of Li-ion batteries, such as the safety guarantee, useful life extension, and charging efficiency improvement [3,4]. Among these problems, how to achieve fast and efficient charge is a key but challenging issue in energy conversion and management of EVs. A well-designed charging strategy can protect batteries from overheat, prolong batteries' cycle life and improve energy conversion efficiency [5,6]. Moreover, inevitable economic losses during the operation of EVs can be reduced by using suitable battery charging protocols [7].

Many approaches have been proposed for battery charging management in recent years. One standard charging method for Li-ion batteries is constant-current constant-voltage (CCCV) [8]. Due to the advantages of simplicity and low-cost implementation, CCCV has become a well-known predefined charging profile in the literature. In addition to this, several variant charging profiles with different predefined forms, such as multistage constant-current (MCC) [9] and boosting charging [10], have been also widely applied to charge Li-ion batteries of EVs. For these charging methods, factors such as current rate and voltage threshold dramatically impact the charging performance. Therefore, it is vital to identify optimize such critical factors and develop optimal charging profiles for various EV applications.

To address this problem, many efforts have been made to improve battery charging performance with various charging objectives. One key objective is the charging speed for Li-ion batteries, which plays a dominated role in determining the acceptance of EVs by customers.

* Corresponding authors.

E-mail addresses: kliu02@qub.ac.uk (K. Liu), xiaosonghu@ieee.org (X. Hu), zyang07@qub.ac.uk (Z. Yang), claudexie@cqu.edu.cn (Y. Xie), sz.feng@siat.ac.cn (S. Feng).

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Nomenclature

I	battery current	E_a	activation energy
V	battery terminal voltage	R_g	universal gas constant
$R_1 C_1$	first resistance-capacity pair	α, β, η, z	dependences of stress factors
$R_2 C_2$	second resistance-capacity pair	B_{new}	cost of new battery
V_1	$R_1 C_1$ network voltage	B_{used}	cost of used battery
V_2	$R_2 C_2$ network voltage	r_{dep}	annual depreciation rate
$OCV(soc)$	battery open circuit voltage versus SOC	DR	discount rate
R_0	battery internal resistance	$JMCC_{BCT}$	charging time objective function
soc	battery state of charge	Δt	sampling time period
C_n	battery capacity	tCC_N	total number of sample points
T_c	battery internal temperature	$JMCC_{BAT}$	average temperature objective function
T_s	battery surface temperature	$JMCC_{BCC}$	charging cost objective function
T_{amb}	ambient temperature	$JMCC_{BEC}$	electricity-loss cost objective function
C_c	battery internal heat capacity	$JMCC_{BAC}$	aging cost objective function
C_s	battery surface heat capacity	tCC_N	sample point when the N th CC phase is end
R_u	heat convection resistance	a	TOU price
R_c	heat conduction resistance	N_{BAC}	total charging cycle number
Q	battery heat dissipation	E_{Ah}	charge throughput for one cycle
T_a	battery radial average temperature	T_{Ah}	charge throughput for all cycles
C_{nloss}	capacity loss	T	time instant of a day
$C_n(initial)$	initial capacity	s_I	initial SOC for charging
$C_n(aged)$	aged capacity	s_T	target SOC for charging
Ct	charge throughput	I_{min}	minimum current for charging
I	average charging current	I_{max}	maximum current for charging
soc	average SOC	V_{min}	minimum terminal voltage
T_a	average battery temperature	V_{max}	maximum terminal voltage
		T_{min}	minimum battery temperature
		T_{max}	maximum battery temperature

Various intelligent algorithms including fuzzy-logic control [11,12], model predictive control [13–15], and heuristic methods [16,17] have been applied to devise proper charging protocols, with the main purpose of decreasing charging time. For these methods, although the charging speed can be enhanced to some extent, there is still a challenge to select proper parameters of intelligent algorithms. Several alternative solutions were reported to consider the battery thermal variation as another crucial objective during charging management [18–21]. Without using appropriate thermal management strategies, the temperature of Li-ion batteries may rise beyond the threshold in several high power applications with large charging current rate, and such overheat would result in substantial battery degradation and potential safety problems, such as fire and explosion [22–24].

Some researchers have also given direct insights into battery health-aware charging, with the overarching goal of prolonging battery service life [27–33]. A key problem for battery health-conscious charging is how to consider the health-related elements into the charging design process [25,26]. On the basis of electrochemical models, several dynamics such as lithium plating, active material loss, and solid electrolyte interface (SEI) film growth, which potentially lead to the battery degradation, can be captured. Corresponding health-conscious charging strategies would then be designed to limit these adverse elements and further restrain battery degradation process. For instance, based upon a reduced electrochemical-thermal model, Perez et al. [27] formulated an optimal battery charging control problem, where the Legendre-Gauss-Radau approach was applied to achieve health-conscious charging subject to phase concentrations and temperature constraints. In [28], a closed loop observer was constructed to capture the lithium deposition state based on a control-oriented electrochemical model. Then a non-destructive charging strategy was proposed to rapidly charge battery while ensuring the satisfactions of health-related constraints. Zou et al. [29] proposed a moving horizon estimation approach to monitor battery electrochemical states, then a fast charging strategy was developed to decrease charging time and protect battery health. Although the critical health-related information inside a battery can be captured by

using a battery electrochemical model, it is still a challenging task to implement these charging strategies in real-world applications due to the requirements of significant computational resources. To alleviate the computational expense arisen from electrochemical models, equivalent electric circuit models (EECMs) have also been adopted in the field of battery charging management with suitable multi-objective optimization techniques. For instance, Fang et al. [30] formulated a linear quadratic control framework based on a second-order resistance-capacitance (RC) model, then a set of model-based charging management strategies was proposed to satisfy user-involved objectives with awareness of battery health. In [31], a dual-objective optimization problem considering the battery health and charging speed was formulated based on the general EECM, where the suitable maximum values of terminal voltage and current were searched to achieve the desired battery service life, while attaining the fastest possible charging speed. Liu et al. [32] formulated three cost functions to consider battery health, charging time and energy conversion during charging, then a heuristic approach named ensemble multi-objective biogeography-based optimization was applied to derive suitable charging patterns. Based upon a coupled electrothermal-aging model, a multi-objective charging optimization problem was formulated to balance the battery charging time and health degradation through using the pseudo-spectral technique in [33]. Even though the EECM-based strategy appears to be promising for battery health-related charging management due to its simple structure and ease to configure parameters, it is still required to simultaneously take other crucial charging objectives and hard constraints into account, to further avoid sub-optimal charging behaviours.

In some other studies, charging cost is also a crucial objective. Currently, most economic charging strategies in EV applications are proposed to alleviate adverse effects on the electricity system such as the power grid, whereas few focus on directly designing charging profiles to benefit cost savings of EV and battery owners [34–36]. For example, in [37,38], the economic cost trade-off between power fade and battery degradation has been performed by designing the charging management strategy in a plug-in hybrid EV. However, only the

constant C-rate charging was considered. From the economic and ecological points of view, five charging strategies with the key inputs of charging curve and energy consumption were designed to limit battery aging while achieving high mileage in [39]. Crow et al. [40] put forward a dual-objective optimization framework which simultaneously takes the valley filling and charging cost into account, then a moving horizon based real-time charging scheme was proposed to balance the requirements of grid-operator and customer jointly. In [41], based upon a battery swap pricing model from the system perspective, a decentralized charging strategy was proposed to enhance the economy-saving and emission-decreasing performance of electric taxis in China. It should be known that unsuitable charging approaches will also result in dramatic economic loss caused by rapid capacity degradation from the perspective of batteries, which cannot be neglected due to the fact that the price of batteries in EVs is nearly half of an EV price [42,43]. Besides, hundreds or thousands of battery cells require to be recharged to support enough power for an EV every day. The economic cost caused by the electrical energy loss during charging is also an inescapable part of the total cost in real-world EV operations. Therefore, it is crucial to develop suitable charging management approaches by simultaneously considering the economic costs of electrical energy loss and battery degradation from the viewpoints of EV and battery owners.

To address the above-mentioned challenges, a novel multi-objective optimization framework is proposed in this paper to optimize charging profiles for achieving economy-conscious Li-ion battery charging management. Several original contributions are highlighted as follows: 1) Crucial and comprehensive charging objectives in terms of the charging speed, average battery temperature, and particularly economic charging cost are formulated for the optimization purpose of charging profiles. These three objectives are subject to a coupled electrothermal-aging model, economic price models, and operated constraints. 2) From the economic charging perspective, both the battery degradation cost and electricity-loss cost are used to generate the charging cost function, where the effects of battery capacity degradation and electrical energy loss are holistically considered in searching the optimal charging profiles. 3) A multi-objective optimization framework associated with the non-dominated sorting genetic algorithm-II (NSGA-II) method is developed to model and solve the highly nonlinear and strongly coupling optimization problem, through which the MCC profile can be optimized to balance various crucial but contradictory objectives during battery charging. 4) Effects of several sensitive parameters including the cut-off voltage, convection resistance, and ambient temperature on the charging performance are comprehensively analysed and discussed based on the obtained Pareto frontiers. 5) For different charging priorities, the presented charging management approach is capable of offering reliable economy-conscious charging with satisfactory equilibrations between charging speed and thermal variations during charging. This is for the first time to simultaneously consider the costs of both electrical energy loss and battery aging in the context of directly designing battery charging profiles. This approach is also applicable to energy management for other types of batteries with proper modelling.

The structure of this paper is listed as follows: first, in Section 2, details of the coupled battery electrothermal-aging model and the economic price models are presented. Based on these models, the pre-defined charging profile and the crucial charging objective functions are formulated, followed by a detailed description of the optimization procedure in Section 3. In Section 4, optimization results including key sensitive parameters and charging cases with different priorities are analysed based on the Pareto frontiers. Finally, conclusions are given in Section 5.

2. Battery model and economic price model

In this study, a coupled electrothermal-aging model is presented to capture the dynamics of battery charging firstly, followed by using the

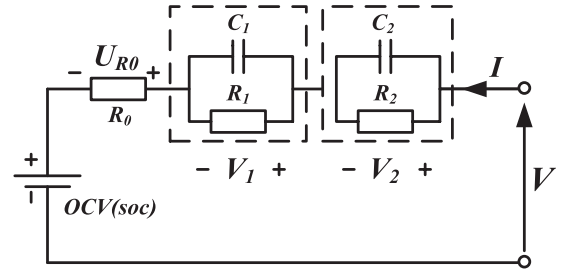


Fig. 1. The Second-order EECM to capture the battery electric behaviour.

economic price models to capture the charging costs caused by the battery degradation and electricity loss. All parameters in these models have been calibrated based on the corresponding experiments.

2.1. Coupled battery model

This high-fidelity coupled battery model is composed of two sub-models: an electrothermal sub-model to simultaneously capture battery electrical and thermal behaviours in a short timescale, and a capacity loss sub-model to predict battery aging dynamics in a cycle-level timescale.

2.1.1. Electrothermal sub-model

During the battery charging process, both the electrical and thermal dynamics influence the battery performance and thereby require to be monitored in real time. For the electrical part, the EECM consisting of some resistances, resistance-capacity (RC) pairs, and open circuit voltage (OCV) source, are widely adopted to describe battery electrical behaviour in many industrial applications [44,45]. In this study, a second-order RC EECM, as shown in Fig. 1, is applied to capture the battery behaviour including the current, voltage, and SOC during a charging cycle.

The second-order EECM is composed of an internal resistance R_0 , two RC pairs (R_1C_1 , R_2C_2), and an ideal battery OCV source. In the figure, I denotes the battery current that is positive for charging, V represents the battery terminal voltage, V_1 is the voltage of the R_1C_1 pair to describe the battery charging transfer process, and V_2 denotes the voltage of the R_2C_2 pair to reproduce the battery diffusion process. Moreover, $OCV(soc)$ represents that the battery open circuit voltage owns a nonlinear relation with the battery SOC. The corresponding equations to capture the electrical behaviour are expressed as:

$$\begin{cases} dsoc(t)/dt = I(t)/C_n \\ dV_1(t)/dt = V_1(t)/R_1(t)/C_1(t) + I(t)/C_1(t) \\ dV_2(t)/dt = V_2(t)/R_2(t)/C_2(t) + I(t)/C_2(t) \end{cases} \quad (1)$$

$$V(t) = OCV(soc(t)) + V_1(t) + V_2(t) + I(t)R_0(t) \quad (2)$$

where C_n stands for the battery capacity with the unit As ($1As = 1/3600Ah$).

For the thermal part, assuming that heat conduction is the only way to transfer heat outside and within the battery, based upon the energy conservation principle, the battery surface temperature T_s and internal temperature T_c at time instant t can be described as follows [49]:

$$\begin{cases} C_s dT_s(t)/dt = (T_{amb} - T_s(t))/R_u + (T_c(t) - T_s(t))/R_c \\ C_c dT_c(t)/dt = (T_s(t) - T_c(t))/R_c + Q(t) \end{cases} \quad (3)$$

where T_{amb} represents the ambient temperature, R_u is a resistance standing for the battery heat convection, R_c is another resistance depicting the battery heat conduction, C_s and C_c represent the battery surface heat capacity and internal heat capacity, respectively. The heat generation $Q(t)$ can be calculated as [46]:

$$Q(t) = I(t)T_c(t)dOCV(t)/dT_c(t) + I(t)(V(t) - OCV(t)) \quad (4)$$

Here the first term on the right side of (4) denotes the heat caused by the entropy change. The second term stands for the Joule's heating. Detailed descriptions of these heating types can be found in [46,47]. The radial average temperature T_a for the Li-ion battery at the time instant t is mathematically calculated as:

$$T_a(t) = (T_c(t) + T_s(t))/2 \quad (5)$$

According to the EECM and thermal sub-model, the battery electrical dynamics are strongly coupled with the battery thermal behaviour. To be specific, at the time instant t , the resistances $R_0(t)$ is mainly affected by the battery internal temperature $T_c(t)$, RC pairs including $R_1(t)C_1(t)$ and $R_2(t)C_2(t)$ are time-varying parameters and mainly associated with $soc(t)$ and $T_f(t)$. Besides, battery electrical dynamics also affect the heat generation $Q(t)$ during charging.

In this study, cylindrical A123 26,650 Li-ion batteries, each with 3.3 V and 2.5 Ah nominal voltage and capacity, respectively, are adopted. Several key characteristics for this type of Li-ion battery are shown in Table 1. The A123 battery can offer long cycle life with a small impedance increase over the lifetime [48]. The thermal parameters for this battery have been identified as the constant values and shown in Table 2 [49]. The detailed calibration process for other time-varying parameters of the battery EECM part and the corresponding identification results can also be found in [49], which are not repeated here due to space limitations.

2.1.2. Capacity loss sub-model

For the aging part of Li-ion battery during charging, we mainly focus on the cycle life rather than the calendar life. In this study, a capacity loss model [50], which has been proved to own the powerful ability to capture the impacts of charging operations on the accumulated aging dynamics, is adopted for the A123 Li-ion battery.

The battery degradation state is defined by the normalized capacity loss as:

$$C_{nloss} = \frac{C_n(initial) - C_n(aged)}{C_n(initial)} \times 100\% \quad (6)$$

where C_{nloss} is the relative capacity loss of battery in percentage, $C_n(initial)$ denotes the initial capacity of new battery, and it is generally equal to the battery nominal capacity. Furthermore, $C_n(aged)$ stands for the capacity of aged battery. Then the battery capacity loss can be described as [46]:

$$C_{nloss} = \sigma(f)Ct^z \quad (7)$$

where f represents several key elements to cause battery aging, Ct is the accumulated current throughput during battery operations, z is a power law parameter to stand for the dependence of current throughput, and $\sigma(f)$ denotes the nonlinear relationship between key elements and battery aging phenomena. After using the average charging current rate I , average SOC soc , and average battery temperature T_a from three cycling cases to validate this capacity loss model, a satisfactory result has been achieved in [46]. Then $\sigma(f)$ can be further expressed as:

$$\sigma(f) = [\alpha soc + \beta] \exp \left[\frac{-E_a + \eta I}{R_g(273.15 + T_a)} \right] \quad (8)$$

where α and β define the dependence on the battery SOC. In order to achieve a satisfactory fitting result, the values of α and β are separated by 0.45 soc into two pairs, as illustrated in Table 3. Other parameters in Eq. (8) are specified in Table 4 for the A123 battery. All these parameters are identified over the experimental data. More detailed information regarding the parameters identification of this capacity loss model can be referred to [50].

Due to the simple structure and ability to couple battery electrochemical mechanisms such as the Arrhenius law, this type of capacity loss model can be easily implemented in real applications such as the

design of battery charging strategy [33]. It should be noted that in Eq. (8), the units of soc and T_a are in fraction and °C, respectively. Battery depletion phenomena are also strongly coupled with the battery electrical and thermal dynamics. The battery aging states especially for the capacity loss and the available charging cycle number will be predicted based on this aging sub-model.

2.1.3. Coupled model

Based on the obtained electrothermal sub-model and capacity loss sub-model, a coupled battery electrothermal-aging model can be formulated. For one charging cycle operation with the fast timescale t , the input vector $u(t)$, state vector $x(t)$, and output vector $y(t)$ for the electrothermal sub-model part can be expressed as

$$\begin{cases} u(t) = I(t) \\ x(t) = [soc(t), V_1(t), V_2(t), T_c(t), T_s(t)]^T \\ y(t) = [soc(t), V(t), T_a(t)]^T \end{cases} \quad (9)$$

Given an injected charging profile and the known initial conditions, the battery dynamics including the battery SOC, terminal voltage, and temperature in a charging cycle can be all captured recursively. For a cycle level timescale with the time instant τ , after obtaining all vectors in Eq. (9) by using the electrothermal sub-model, $I(\tau)$, $soc(\tau)$, $T_a(\tau)$ and $Ah(\tau)$ in the τ -th charging cycle can be calculated and further injected into Eqs. (7) and (8) to predict the degraded capacity of the battery caused by this specific charging profile. The corresponding input vector $u_{bl}(\tau)$ and output vector $y_{bl}(\tau)$ for the capacity loss sub-model part are presented as

$$\begin{cases} u_{bl}(\tau) = [I(\tau), soc(\tau), T_a(\tau), Ct(\tau)]^T \\ y_{bl}(\tau) = C_{nloss}(I(\tau), soc(\tau), T_a(\tau), Ct(\tau)) \end{cases} \quad (10)$$

In this way, the electrical and thermal behaviours in a charging cycle can be captured with the fast timescale t , while the battery aging dynamics in the cycle level timescale τ can be also predicted.

2.2. Economic price model

For Li-ion battery charging, one major economic cost is the cost of battery degradation. It should be known that the battery capacity will degrade during its service life. When a battery capacity degrades to 80% of its original nominal value, this battery is no longer competent for EV applications. In order to evaluate this economic cost caused by the battery capacity degradation during charging, a proper model to describe the battery cost during operations is necessary. Given that a used Li-ion battery can be resold, a battery resale cost model [51] defined in Eq. (11), is applied to estimate the equivalent cost of battery aging in this study.

$$B_{used} = \frac{B_{new} \times (1 - r_{dep})^{T_{use}}}{(1 + DR)^{T_{use}-1}} \quad (11)$$

where T_{use} represents the lifetime of a battery in EV applications, B_{new} and B_{used} stand for the costs of new battery and used battery, respectively, r_{dep} denotes the annual depreciation rate, DR is the discount rate. The units of both r_{dep} and DR are set in percentage. Here, the cost of a

Table 1
Characteristics for the A123 26,650 LiFePO₄ battery [48].

Parameters	Specifications	Values	Units
C_n	Cell capacity (nominal)	2.5	Ah
$h \times \phi$	Cell dimensions	26×65	mm
V_{nom}	Nominal voltage	3.3	V
R_{im}	Internal impedance (1 KHz)	6	mΩ
L_B	Cycle life at 10C discharge	>1000	cycles
T_{sto}	Temperature for storage	−40 to 60	°C
T_{ope}	Temperature for operation	−30 to 50	°C

Table 2
Parameters for the A123 26650 LiFePO₄ battery thermal model [49].

Parameters	Descriptions	Values	Units
C_c	Internal heat capacity	4.5	JK^{-1}
C_s	Surface heat capacity	62.8	JK^{-1}
R_c	Heat conduction resistance	3.08	KW^{-1}
R_u	Convection resistance	1.93	KW^{-1}
T_{amb}	Ambient temperature	25	$^{\circ}C$

Table 3
Values for α and β at different SOC levels [46].

Parameters	$soc < 0.45$	$soc \geq 0.45$
α	2897.8	2694.3
β	7413.1	6025.6

Table 4
Results for constant parameters in battery aging model [46].

Parameters	Specifications	Values	Units
E_a	Activation energy	31,500	$Jmol^{-1}$
R_g	Universal gas constant	8.314	$Jmol^{-1}$
η	Battery current dependence	152.6	–
z	Current throughput dependence	0.57	–

new A123 LiFePO₄ battery is set to 35 Chinese Yuan (CNY) each cell, recommended by the open market.

Another primary economic cost during charging is the cost of electricity loss, due to the fact that it is impossible to convert all electrical energy into the effective chemical energy during charging. In fact, some thermal reactions will cause the loss of electrical energy in the form of waste heat. Therefore, it is necessary to select a suitable model or function to describe the electricity cost for battery charging operations. The peak-valley time-of-use (TOU) price has been justified as a common way around the world to describe the electricity price during a day. In this study, the residential Beijing peak-valley time-of-use (TOU) price from [36] is adopted as the electricity price, where the histogram of this TOU price and details are shown in Fig. 2 and Table 5, respectively.

3. Formulations of charging profile and crucial objectives

Fig. 3 illustrates the overall multi-objective optimization framework for economy-conscious charging. Given the battery coupled model and the economic price model, crucial objectives including the charging time, average temperature, and particularly charging cost are formulated. Then the economy-conscious battery charging can be achieved with suitable energy and time management.

3.1. MCC profile

In this research, the multistage constant-current (MCC) charging profile is optimized. This MCC charging profile can be divided into several constant-current (CC) phases, as shown in Fig. 4. Since most Li-ion batteries are less susceptible to lithium plating at low SOC levels, a relatively large constant current rate I_{CC1} can be adopted at the beginning of the MCC charging process until enough energy throughput is transferred into the battery. After that, a series of stepwise decreased CC phases will be implemented until the last CC phase is finished with the constant current rate I_{CCN} . One CC phase will turn into the next CC phase when the battery terminal voltage reaches a predefined cut-off voltage V_{cut} .

Although the MCC profile is easy to be implemented in EV

applications, the key and challenging problem is to set the proper values of charging currents and voltage for the entire charging process. The charging time of the MCC profile is primarily determined by the cut-off voltage V_{cut} and the corresponding current rates in each CC phase ($I_{CC1}, I_{CC2}, \dots, I_{CCN}$). According to the datasheet from Li-ion battery manufacturers, V_{cut} is usually set to the maximum voltage level with the purpose of improving the capacity utilization. For the charging current, a large current rate can speed up the charging process, but also result in severe problems such as lithium plating, large energy loss, and over-heat, further to affect the battery cycle life significantly. Then the economic costs for the wasted electrical energy and degraded battery capacity would be increased. Therefore, it is vital to search the suitable current rates in the MCC profile for efficient equilibrations among the charging time, battery temperature, and particularly economic charging costs.

3.2. Charging objectives

According to the battery coupled model and economic price models, several crucial but contradictory objectives are formulated to achieve economy-conscious charging management.

For the battery charging time (BCT), less charging time means that the same battery charging process can be finished with a faster speed. In the MCC profile, the objective function for BCT is described as:

$$JMCC_{BCT} = \Delta t * tCC_N / 60 \quad (12)$$

where Δt is the sampling time period, tCC_N represents the total number of sample points when the battery is charged from the initial SOC to the final target SOC level and $JMCC_{BCT}$ is with the unit minutes (M).

For the battery average temperature (BAT), less average temperature can protect the battery from overheating under the same ambient temperature. In the MCC profile, the objective function $JMCC_{BAT}$ for BAT can be expressed as:

$$JMCC_{BAT} = \Delta t * \left[\sum_{t=0}^{tCC_1} T_a(t) + \dots + \sum_{tCC_{i+1}}^{tCC_2} T_a(t) + \sum_{tCC_{N-1+1}}^{tCC_N} T_a(t) \right] / tCC_N \quad (13)$$

where $tCC_1, tCC_2, \dots, tCC_N$ represent the sample points when each CC phase is ended, respectively, $T_a(t)$ is calculated by Eq. (5) to denote the radial average temperature of the battery at the time instant t .

For the battery charging cost (BCC) in this study, the total economic cost can be divided into two types: battery electricity-loss cost (BEC) and battery aging cost (BAC). In the MCC profile, the objective function $JMCC_{BCC}$ for BCC is described as:

$$JMCC_{BCC} = JMCC_{BEC} + JMCC_{BAC} \quad (14)$$

where $JMCC_{BEC}$ represents the cost caused by the electricity-loss during charging and can be expressed as:

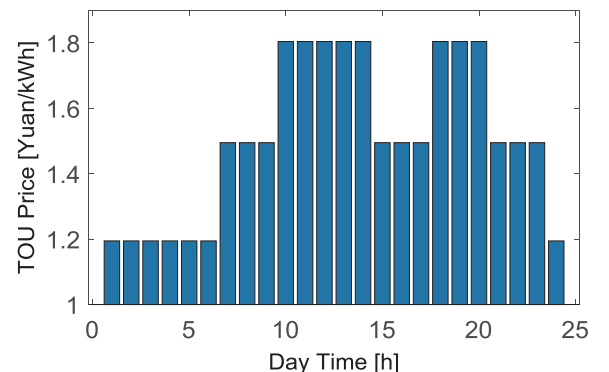


Fig. 2. The histogram of Beijing peak-valley TOU price.

Table 5
Detailed TOU price for the Beijing peak-valley.

Pattern	Time Period	Price [Yuan/kWh]
On-peak	10:00–15:00	1.8044
	18:00–21:00	
Shoulder	7:00–10:00	1.4950
	15:00–18:00	
Off-peak	21:00–23:00	1.1946
	23:00–7:00	

$$\begin{cases} JMCC_{BEC} = \Delta t * \left[\sum_{t=0}^{t_{CC1}} f_{EC}(t) + \dots + \sum_{t_{CC1}+1}^{t_{CC2}} f_{EC}(t) + \dots + \sum_{t_{CCN-1}+1}^{t_{CCN}} f_{EC}(t) \right] \\ f_{EC}(t) = a(t) * [I^2(t)R_0(t) + V_1^2(t)/R_1(t) + V_2^2(t)/R_2(t)] \end{cases} \quad (15)$$

where $f_{EC}(t)$ stands for the cost of electrical energy loss occurred at the time instant t during battery charging, $a(t)$ is the TOU price at the time instant t . The value of $a(t)$ would remain unchanged for a long time-scale and is affected by the time instant T of a day. Detailed relationships of $a(t)$ and T are described as:

$$a(t) = f(t \in T) = \begin{cases} \frac{1.1946}{3.6 \times 10^6} & T \in (23:00 - 7:00) \\ \frac{1.4950}{3.6 \times 10^6} & T \in (7:00 - 10:00) \text{ or } T \in (15:00 - 18:00) \text{ or } T \in (21:00 - 23:00) \\ \frac{1.8044}{3.6 \times 10^6} & T \in (10:00 - 15:00) \text{ or } T \in (18:00 - 21:00). \end{cases} \quad (16)$$

For the cost of battery degraded capacity during charging, the objective function $JMCC_{BAC}$ is calculated by:

$$\begin{cases} JMCC_{BAC} = \frac{B_{new} - B_{used}}{N_{BAC}} \\ N_{BAC} = \frac{T_{Ah}}{E_{Ah}} \end{cases} \quad (17)$$

where B_{used} can be obtained by the battery resale cost model in Eq. (11), N_{BAC} stands for the total charging cycle number when the battery capacity decays to its end of life (EOL). Moreover, T_{Ah} and E_{Ah} represent the total throughput in Ah for all charging cycles and one cycle, respectively. For EV operations, the EOL of Li-ion battery is usually achieved when the battery capacity degrades to 80% of its nominal

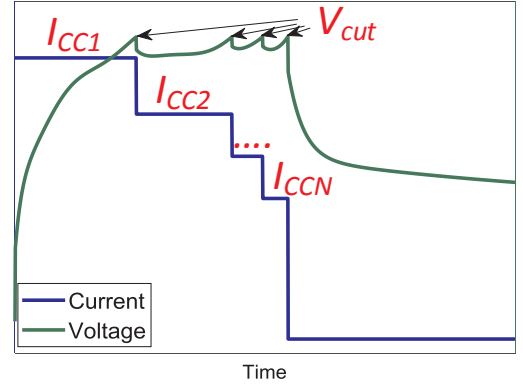


Fig. 4. Charging current and terminal voltage for MCC profile.

value. Assuming that the battery is charged by the same MCC profile with the same conditions in each charging cycle, T_{Ah} is mainly affected by the I , soc , and T_a of this specific MCC profile. Then T_{Ah} can be calculated as:

$$T_{Ah} = \left[\frac{20}{\sigma_{f_{unct}}(I, soc, T_a)} \right]^{1/2} \quad (18)$$

$$\begin{cases} I = \sum_{t=0}^{t_{CCN}} I(t) / t_{CCN} \\ soc = \sum_{t=0}^{t_{CCN}} soc(t) / t_{CCN} \\ T_a = \sum_{t=0}^{t_{CCN}} T_a(t) / t_{CCN}. \end{cases} \quad (19)$$

For the E_{Ah} of the specific MCC profile, the charging throughput is also an aggregated value as:

$$E_{Ah} = \frac{\Delta t}{3600} \sum_{t=0}^{t_{CCN}} I(t). \quad (20)$$

In this study, once the current rates in each CC phase of the MCC profile are optimized, all elements of the cost functions $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$ can be obtained by the coupled battery model and the economic price model. Namely, current rates I_{CC1} , I_{CC2} , ..., I_{CCN} are decisive variables that need to be carefully optimized for the economy-conscious charging.

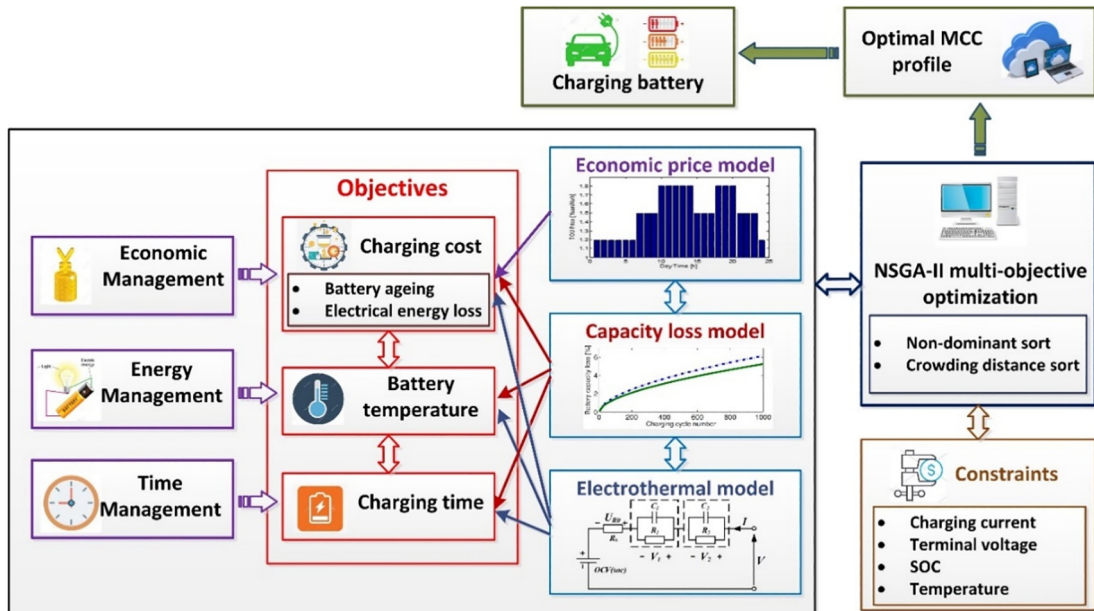


Fig. 3. Overall multi-objective optimization framework for economic-conscious charging.

3.3. Optimization procedure

The optimization of the MCC profile is a highly nonlinear and strong-coupling process. Many time-varying parameters such as capacitors and resistances in the battery coupled model are strongly associated with the temperature and SOC levels. Battery capacity degradation is also significantly affected by the electrical and thermal dynamics of the battery. Besides, many constraints such as the charging current, terminal voltage, battery SOC, and temperature require to be considered during the optimization process as follows:

$$\begin{cases} I_{min} \leq I(t) \leq I_{max} \\ V_{min} \leq V(t) \leq V_{max} \\ s_I \leq soc(t) \leq s_T \\ T_{min} \leq T_a(t) \leq T_{max} \end{cases} \quad (21)$$

For this time-varying and complicated optimization problem, an effective multi-objective optimization technique is required to optimize the MCC profile and equilibrate crucial but contradictory charging objectives. In this regard, meta-heuristic optimization methods are immune from the NP-hard and highly nonlinear problems, providing potential powerful tools for solving the problem [52]. Among various meta-heuristic multi-objective optimizers, NSGA-II has been justified as a powerful approach to solve complex multi-objective optimization problems in many real-world applications, owing to its strong abilities of keeping the elitism optimization and preserving the diversity [53,54]. Therefore, NSGA-II approach is herein employed to search the optimal parameters of the MCC profile, with the target of achieving economy-conscious charging with different charging priorities.

The optimization flow chart by searching the proper MCC profile to equilibrate charging time, battery average temperature, and particularly economic charging cost is shown in Fig. 5. The detailed process is summarized as follows.

Step 1: Set constraints for the battery charging process. Detailed constraints include: 1) the initial and target SOC levels: s_I and s_T ; 2) limits of battery current, voltage, and temperature: I_{min} , I_{max} , V_{min} , V_{max} , T_{min} , and T_{max} .

Step 2: Set parameters for the MCC charging profile. Detailed parameters include: 1) the cut-off voltage: V_{cut} ; 2) the number of CC phases: N_{CC} .

Step 3: Select charging objectives based on the user requirements. According to the priorities for different battery applications, the suitable combination of charging objectives (BCT, BAT, and BCC) is formulated.

Step 4: Initialize the parameters for the NSGA-II approach. Main parameters include: 1) the number of generation: G_m ; 2) the population size: N_p .

Step 5: For $k = 1$ to k_{max} , do

- 1) In each CC phase, calculate the cost functions for the selected charging objectives until the battery SOC goes up to the target level s_T . A CC phase will be terminated and then jump to another CC phase as the battery terminal voltage reaches the cut-off voltage V_{cut} .
- 2) Evaluate the values of $J_{MCC_{BCT}}$, $J_{MCC_{BAT}}$, and $J_{MCC_{BCC}}$ for the whole charging process by summing the cost function terms from each CC phase. Then the performance of the optimized MCC profile can be analysed based on the optimal solution set from the drawn Pareto frontier.
- 3) If the performance of the obtained MCC profile is unsatisfactory, search a new MCC profile by the NSGA-II approach again. This optimization process will be stopped as the end criterion k_{max} is achieved.

By following the proposed framework, the economy-conscious charging can be procured by using the optimized MCC profile. Besides, the charging time and battery average temperature can be also

equilibrated accordingly.

4. Results and discussion

In this Section, sensitivities of key parameters including the cut-off voltage, convection resistance, and ambient temperature are first explored to analyse their effects on the optimization results of the MCC profile. Then the optimal solution set is drawn as the Pareto frontier to equilibrate contradictory objectives for different user demands. Three types of optimized MCC profiles including two extreme charging cases and a balanced charging scheme are conducted with the detailed analysis of their charging trajectories. Detailed comparisons of these MCC profiles are also carried out to investigate their effectiveness on the trade-offs of the charging time, charging cost, and battery average temperature. Besides, the battery capacity degradations of these MCC profiles are also analysed. All the optimization tests are conducted in the Matlab with a 2.40 GHz Intel Pentium 4 CPU. For constant parameters setting, the sampling time period is set as $\Delta t = 1s$. Initial and final target SOC levels are set as $s_I = 0.1$ and $s_T = 0.95$, respectively. The initial temperatures of both battery surface and interior are set equal to the ambient temperature, $T_s(0) = T_I(0) = T_{amb}$. Limits of the battery current, voltage, and temperature during charging are set as: $I_{min} = 0A$, $I_{max} = 10A$, $V_{min} = 3.0V$, $V_{max} = 3.6V$, $T_{min} = 15^\circ C$, and $T_{max} = 45^\circ C$. For the NSGA-II approach, the population size and the number of generation are set to $N_p = 360$ and $G_m = 60$, respectively, for all tests.

4.1. Sensitivity analysis for the MCC profile

For the MCC charging profile, several key parameters such as the

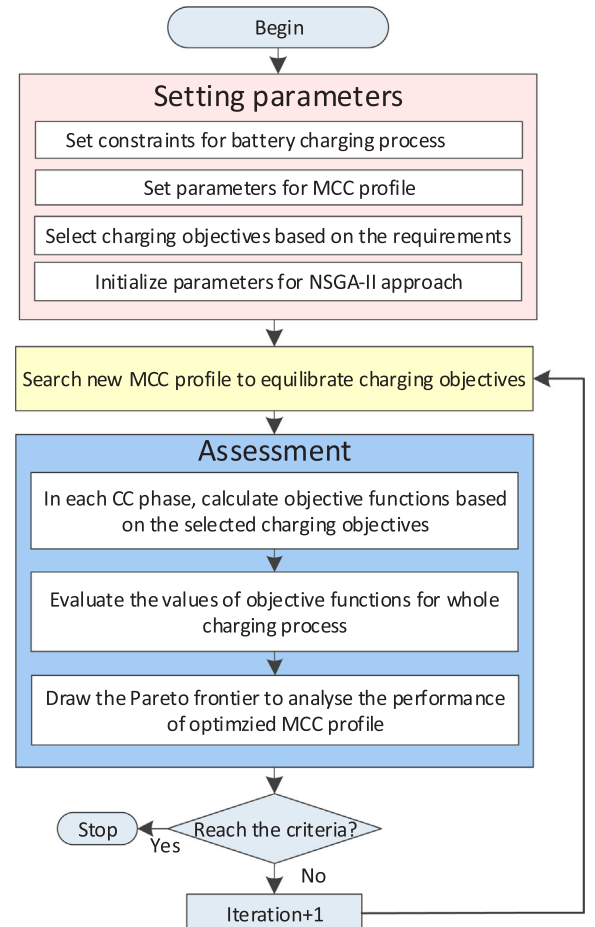


Fig. 5. The optimization flow chart by searching the proper MCC profile to equilibrate charging objectives.

cut-off voltage, convection resistance, and ambient temperature play a vital role in the optimization results for equilibrating charging objectives, which need to be carefully determined. It should be known that for the multi-objective optimizations, the Pareto frontier is capable of providing a set of optimal solutions to graphically illustrate the conditions that one objective function would not be improved without making another objective function worse. If the Pareto frontier for a specific case distributes evenly and widely, and the points in this Pareto frontier get closer to the origin of coordinates (here the origins of Figs. 6–8 are all at bottom), the optimal solutions for such case become better. In the light of this, sensitivities of these three key parameters are analysed in detail via the Pareto frontier to quantify their effects on the charging performance.

4.1.1. Sensitivity of cut-off voltage

The battery cut off voltage V_{cut} is important for the capacity utilization during charging. A small value of V_{cut} restrains the battery capacity utilization, but too large V_{cut} would cause damages or safety hazards to batteries. In this test, the initial current of the first CC phase I_{CC1} is set within 5A (2C) and 10A (4C). The convection resistance and ambient temperature are set to $R_u = 3.08KW^{-1}$ and $T_{amb} = 25^\circ C$, respectively. Six cases of V_{cut} including 3.60 V, 3.58 V, 3.56 V, 3.54 V, 3.52 V, and 3.50 V are conducted to explore the sensitivity of the battery cut-off voltage. Units of $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$ are minute (M), $^\circ C$, and Yuan, respectively.

According to the Pareto frontiers for the MCC profile with different V_{cut} from Fig. 6, it is evident that a better optimal solution set can be achieved by a larger V_{cut} . To be specific, the optimal solution set with $V_{cut} = 3.6V$ contains the largest number of optimal points for all objective functions including $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$. As V_{cut} decreases, the number of obtained optimal points will become less, and the Pareto frontier presents a trend to move to the right, resulting in a fact that more time is required for battery charging. Therefore, a large cut-off voltage is capable of improving the speed for battery charging significantly. Besides, both the values of $JMCC_{BAT}$ and $JMCC_{BCC}$ tend to be reduced with decreased V_{cut} . For example, the minimal values obtained under $V_{cut} = 3.50 V$ are $JMCC_{BAT} = 25.90^\circ C$ and $JMCC_{BCC} = 0.0030 Yuan$, respectively, which are 1.3% and 6.3% less than those under $V_{cut} = 3.60 V$ (here is $JMCC_{BAT} = 26.23^\circ C$ and $JMCC_{BCC} = 0.0032Yuan$, respectively). It is concluded that a small cut-off voltage results in low charging current, hence the battery average temperature rise will be restrained, further to cause less charging cost, but the time of charging is sacrificed accordingly.

4.1.2. Sensitivity of convection resistance

Heat convection resistance R_u is a key parameter to represent the heat convection between the ambient environment and battery shell, which is affected by the thermal management operations, such as air fan or liquid cooling. Natural heat convection is generally with a large value of R_u . For the parameters setting of this test, I_{CC1} is also set within 5A and 10A. The cut-off voltage is set to $V_{cut} = 3.6V$. The ambient temperature during charging is fixed as $T_{amb} = 25^\circ C$. Five cases of R_u including $1KW^{-1}$, $3.08KW^{-1}$, $5KW^{-1}$, $10KW^{-1}$, and $15KW^{-1}$ are performed to analyse the effects of R_u on the optimization results of the MCC profile.

After the optimization, the Pareto frontiers for various heat convection resistances are illustrated in Fig. 7. Given the same I_{CC1} and T_{amb} , different R_u would also result in different trade-off results of $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$. As R_u increases, the Pareto frontier tends to move to the upper. For the battery average temperature and charging cost, increasing heat convection resistance will cause the values of both $JMCC_{BAT}$ and $JMCC_{BCC}$ to become larger. In the case of $R_u = 15KW^{-1}$, the minimal values of $JMCC_{BAT}$ and $JMCC_{BCC}$ are $27.65^\circ C$ and $0.0034Yuan$, respectively, which are over 7.8% and 17.2% more than those to the case of $R_u = 1KW^{-1}$ (here is $JMCC_{BAT} = 25.66^\circ C$, $JMCC_{BCC} = 0.0029Yuan$). On the contrary, the

time of battery charging will reduce when the heat convection resistance is increased. Compared to $R_u = 15KW^{-1}$ with the minimum $JMCC_{BCT}$ as 14.01 M, this value increases to 15.02 M (over 8.6% increase) in the case of $R_u = 1KW^{-1}$. This is mainly because that a large R_u results in the increase of the battery average temperature, further causing the decrease of the battery internal resistance, then the charging time will be shortened accordingly. To summarize, a lower heat convection resistance favours the charging cost and battery average temperature, at a sacrifice of the speed of battery charging.

4.1.3. Sensitivity of ambient temperature

The ambient temperature T_{amb} is also a key parameter for the MCC profile. In this test, I_{CC1} is set initially between 5A and 10A. Other parameters are fixed as: $V_{cut} = 3.6V$, and $R_u = 3.08KW^{-1}$. Six cases of T_{amb} including $15^\circ C$, $20^\circ C$, $25^\circ C$, $30^\circ C$, $35^\circ C$, and $40^\circ C$ are conducted to explore the sensitivity of the ambient temperature.

The effects of the ambient temperature on the optimized MCC profile are illustrated in Fig. 8. Clearly, increasing the ambient temperature would lead the obtained Pareto frontier to move to the left-upper, which means that the time of charging is shortened but both the battery average temperature and charging cost will increase adversely. Quantitatively, the minimum $JMCC_{BCT}$ is just 12.02 M at the highest ambient temperature case ($T_{amb} = 40^\circ C$), which is over 45% less than that at the lowest ambient temperature case $T_{amb} = 15^\circ C$ (here the minimum $JMCC_{BCT}$ increases to 21.9 M). For the battery average temperature and charging cost, the minimum $JMCC_{BAT}$ and $JMCC_{BCC}$ at the condition of $T_{amb} = 15^\circ C$ are just $17.58^\circ C$ and $0.0023Yuan$, respectively. But at the case of $T_{amb} = 40^\circ C$, those values rise up to $43.27^\circ C$ and $0.0114Yuan$, respectively. Therefore, a small ambient temperature favours achieving low charging cost and average temperature rise, but the battery internal resistance will be increased, further inducing a longer battery charging period.

In summary, the optimization results of the MCC profile are critically associated with these key parameters. A large cut-off voltage benefits the charging time. A low heat convection resistance and a small ambient temperature favour both the charging cost and battery average temperature. It is crucial to determine these parameters carefully, based on various requirements from different battery applications.

4.2. Tests for various charging cases

In this subsection, optimization results of various combinations of two charging objectives are analysed firstly. Then three types of MCC charging are described with detailed comparisons to investigate their efficacies on the trade-offs of crucial charging objectives. Here, the initial I_{CC1} is fixed within 5A and 10A. The decrement of current during each CC phase is set between 1.5A and 2.5A. The cut-off voltage is 3.6 V. Other parameters are set to $R_u = 3.08KW^{-1}$ and $T_{amb} = 25^\circ C$, respectively. After the optimization, the Pareto frontiers of different combinations of two charging objectives are illustrated in Fig. 9.

According to the optimization results for dual-objectives by using the MCC profile, the optimal particles in the Pareto frontiers of Fig. 9(a)

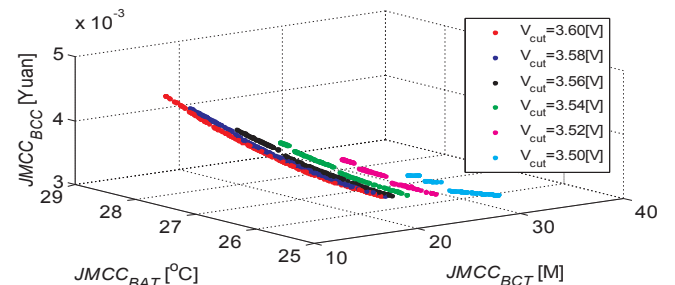


Fig. 6. Effects of cut-off voltages on the optimized MCC profile via Pareto frontier.

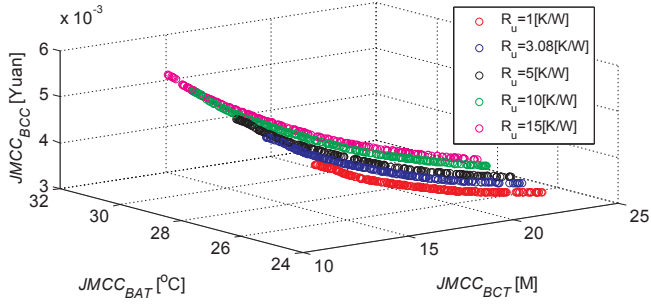


Fig. 7. Effects of convection resistances on the optimized MCC profile via Pareto frontier.

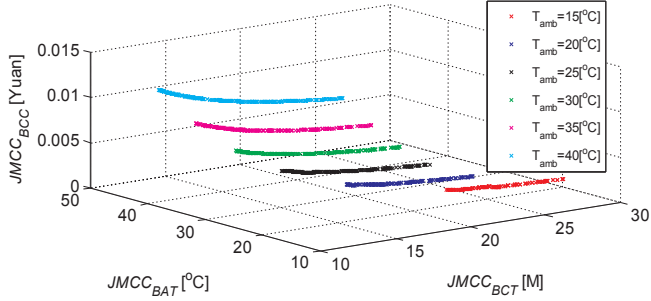


Fig. 8. Effects of the ambient temperature on the optimized MCC profile via Pareto frontier.

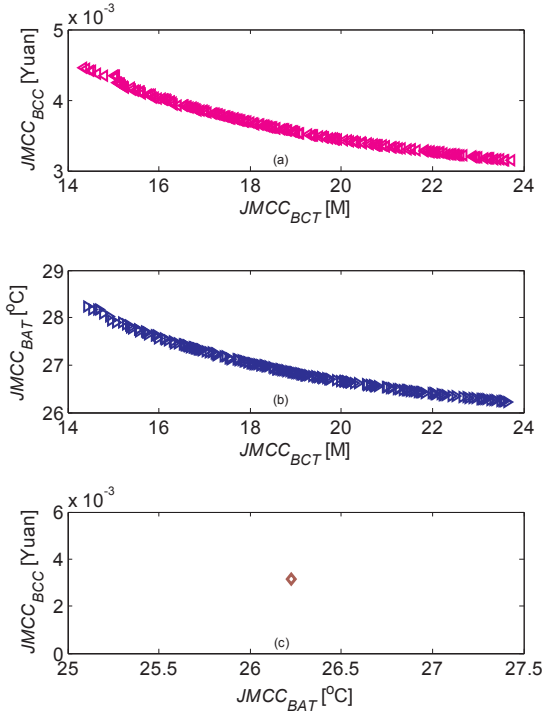


Fig. 9. Optimization results of the MCC profile with two objectives via Pareto frontier.

and (b) are uniformly distributed in the wide regions, which means that both objectives of the charging cost and battery average temperature conflict with the charging time. Clearly, a faster charging speed is achieved by using a larger current rate in the MCC profile, but this increased charging current also causes larger thermal reactions occurred in the battery, hence the battery average temperature will rise up. On the other hand, both the average SOC and average current of the

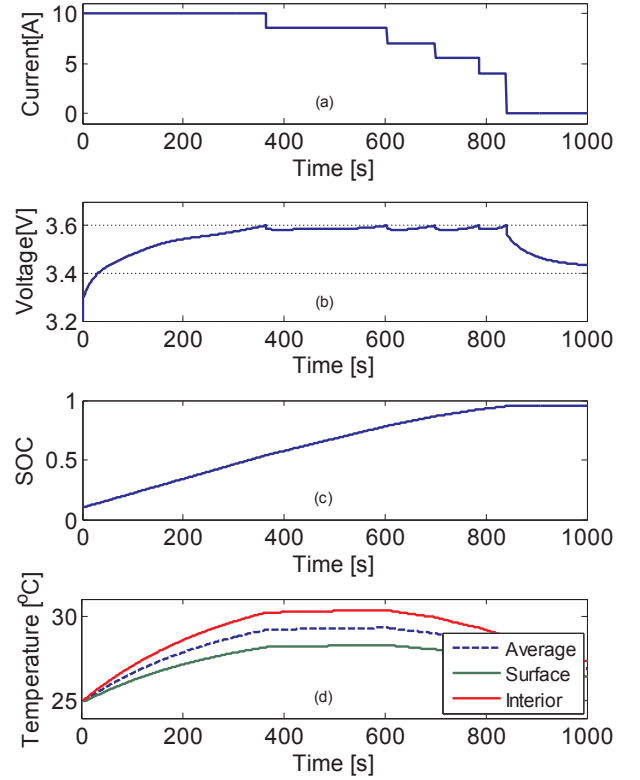


Fig. 10. Detailed MCC trajectories for the minimum time charging.

whole charging process become higher by using a larger current rate, further to aggravate electrical energy loss and battery capacity degradation during charging. Hence, the economic charging cost will be intensified. It is also interesting to note that the optimal particles in the Pareto frontier of Fig. 9(c) finally converge into a single point, which means that there is no conflict between the objectives of charging cost and battery average temperature. Next, two extreme cases from the obtained Pareto frontiers and five balanced charging cases are examined in detail.

4.2.1. Minimum time charging

Fig. 10 shows the optimized charging trajectories including the battery current, voltage, SOC, and temperature for the extreme case of minimum time charging. In order to achieve the fastest charging speed, the maximum constrained constant current 10A is used firstly, causing the battery terminal voltage to rise up rapidly till the upper voltage threshold 3.6 V, followed by four stepwise decreased CC phases. In this case, only 14.02 min is taken for the whole charging process. The current decrements after every CC phase are all optimized to the minimum constraints as 1.5A, leading both the battery SOC and temperature to increase dramatically (here the battery surface temperature rises up to 28.28 °C, while the internal temperature increases to 30.35 °C). It is evident that the largest charging current is required in each CC phase to achieve the fastest charging speed. This case is unfavourable to restrain the battery terminal voltage and temperature rise.

4.2.2. Minimum economic loss charging

Next, another extreme case – minimum economic loss charging is investigated. It should be known that the minimum economic loss and lowest battery temperature rise can be achieved simultaneously during charging, due to the fact that there is no conflict between the economic charging cost and battery average temperature. Detailed trajectories of the MCC profile for this case are illustrated in Fig. 11. Interestingly, just two stepwise decreased CC phases appear in this case. In order to achieve the minimum economic loss, the smallest constrained current

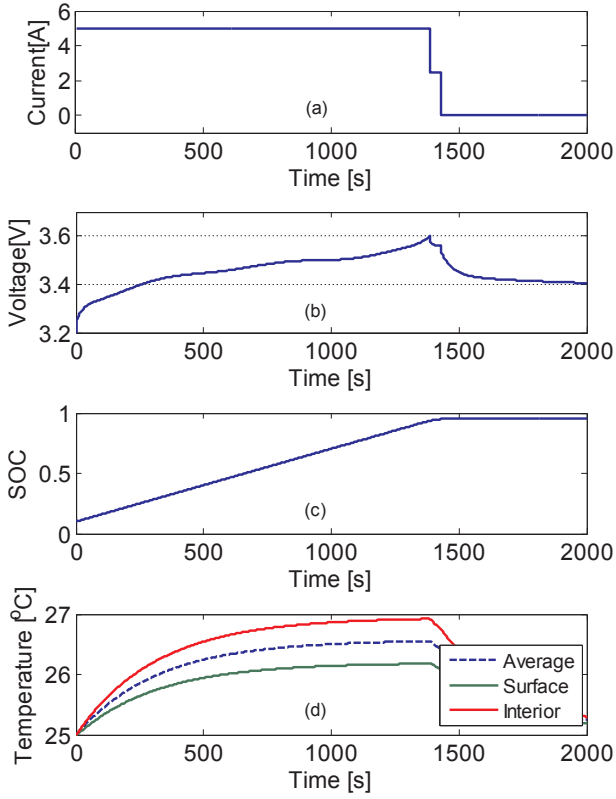


Fig. 11. Detailed MCC trajectories for the minimum economic loss charging.

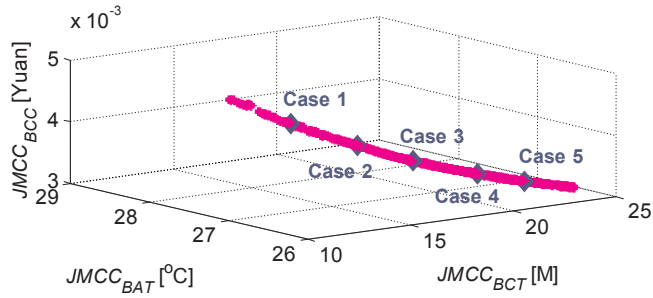


Fig. 12. Optimization results of the MCC profile with three objectives via Pareto frontier.

value (5.01A) is applied initially, causing the terminal voltage to increase slightly during the first CC phase. Another stepwise CC phase with 2.52A charging current is used in a short time period to achieve the final SOC goal. The total charging time in this case becomes 23.82 min, nearly twice more than that from the minimum time charging. But the battery average temperature in this case increases very slowly (here the battery surface and internal temperature increase to 26.18 °C and 26.92 °C, respectively), which can help thwart the battery capacity degradation during charging. Then the economic loss caused by battery aging will be also reduced.

4.2.3. Balanced charging

Apart from two extreme charging cases, the optimization results of balanced charging with three objectives are also plotted in the form of Pareto frontier in Fig. 12. Not surprisingly, objective functions $JMCC_{BAT}$ and $JMCC_{BCC}$ conflict with the function $JMCC_{BCT}$. By sacrificing the battery charging speed, the battery temperature rise and economic loss during charging would be substantially restrained. Five balanced charging cases from the Pareto frontier are exemplified. Case 1 and Case 2 stand for quick charging solutions that take less charging time to

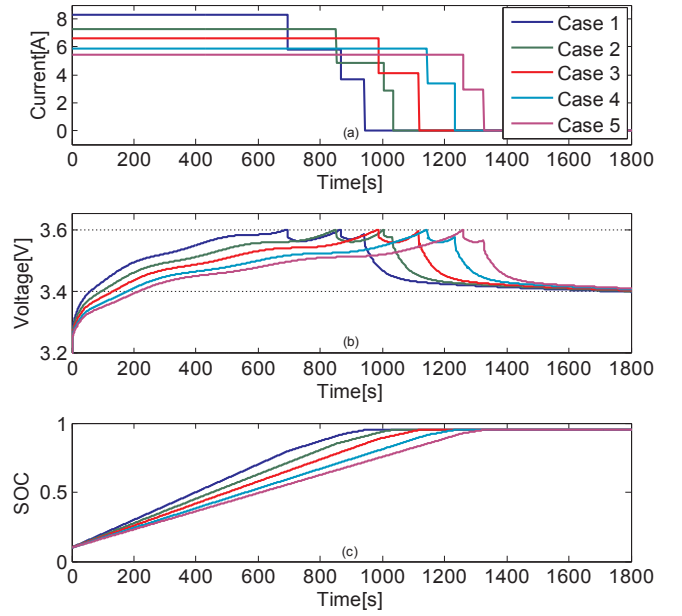


Fig. 13. Electrical dynamics of the MCC profile for the balanced charging: (a) current, (b) voltage, and (c) SOC.

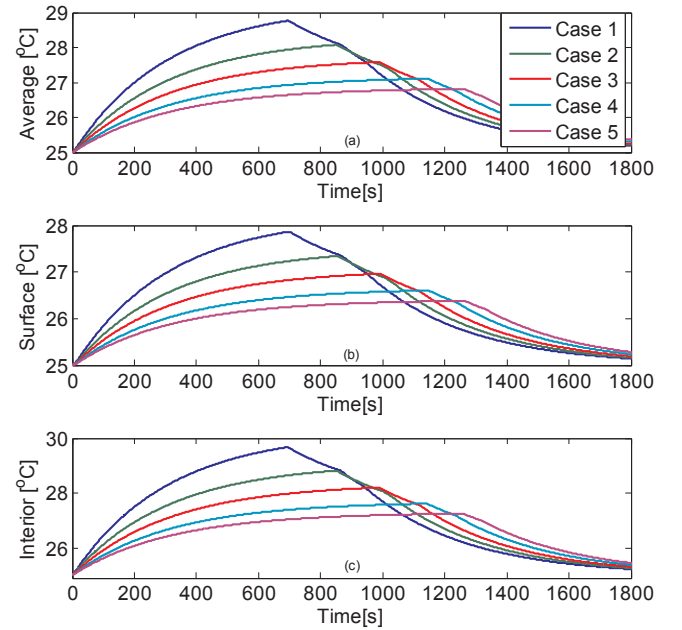


Fig. 14. Thermal dynamics of the MCC profile for the balanced charging: (a) average temperature, (b) surface temperature, and (c) internal temperature.

achieve the SOC target. Case 4 and Case 5 prefer smaller charging cost and lower average temperature rise in the entire charging process. Case 3 is in a neutral position during charging. The electrical dynamics and thermal dynamics of these balanced charging cases are plotted in Figs. 13 and 14, respectively. Detailed charging currents, along with values of objective functions including $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$, are illustrated in Table 6 for all the selected charging cases.

From Figs. 13 and 14, as the required charging time increases from Case 1 to Case 5, the initial charging current in the first CC phase reduces inevitably. The total number of CC phases for the whole MCC charging is also decreased gradually. Both the terminal voltage and battery temperature variation also present decline tendencies in the meantime. Quantitatively, compared with the quick charging solution that takes 15.72 min for the whole charging in Case 1, the required

Table 6Values of the charging currents, $JMCC_{BCT}$, $JMCC_{BAT}$, and $JMCC_{BCC}$ for the selected cases.

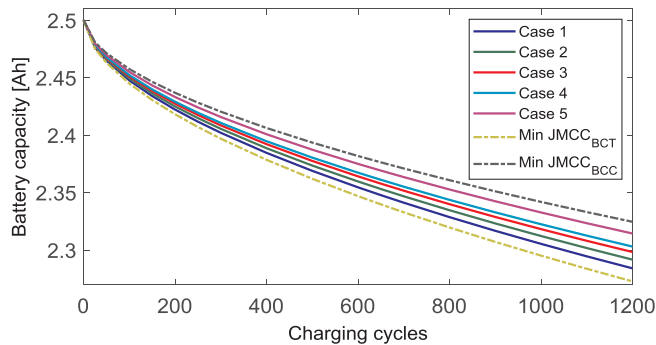
Case study	I_{CC1} [A]	I_{CC2} [A]	I_{CC3} [A]	I_{CC4} [A]	I_{CC5} [A]	$JMCC_{BCT}$ [M]	$JMCC_{BAT}$ [°C]	$JMCC_{BCC}$ [Yuan]
Min $JMCC_{BCT}$	10.00	8.49	7.01	5.52	4.04	14.02	28.36	0.0045
Case 1	8.31	5.82	3.66	–	–	15.72	27.71	0.0041
Case 2	7.31	4.79	2.86	–	–	17.23	27.25	0.0038
Case 3	6.59	4.14	–	–	–	18.63	26.92	0.0036
Case 4	5.89	3.42	–	–	–	20.57	26.61	0.0034
Case 5	5.43	2.95	–	–	–	22.10	26.41	0.0033
Min $JMCC_{BCC}$	5.01	2.52	–	–	–	23.83	26.22	0.0031

Table 7Detailed values of the $JMCC_{BAT}$ and $JMCC_{BCC}$ for the selected cases.

Case study	$JMCC_{BAT}$ [°C]	$MaxT_a$ [°C]	$MaxT_s$ [°C]	$MaxT_l$ [°C]	$JMCC_{BCC}$ [Yuan]	$JMCC_{BAC}$ [Yuan]	$JMCC_{BEC}$ [Yuan]
Min $JMCC_{BCT}$	28.36	29.31	28.28	30.35	0.0045	0.0038	7.670e-4
Case 1	27.71	28.77	27.86	29.68	0.0041	0.0034	6.926e-4
Case 2	27.25	28.08	27.34	28.82	0.0038	0.0032	6.348e-4
Case 3	26.92	27.58	26.96	28.20	0.0036	0.0030	5.879e-4
Case 4	26.61	27.11	26.60	27.61	0.0034	0.0029	5.420e-4
Case 5	26.41	26.82	26.38	27.25	0.0033	0.0028	5.102e-4
Min $JMCC_{BCC}$	26.22	26.55	26.18	26.92	0.0031	0.0027	4.786e-4

Table 8Detailed values of the $JMCC_{BAC}$ for the selected cases.

Case study	$JMCC_{BAC}$ [Yuan]		
	On-peak	Shoulder	Off-peak
Min $JMCC_{BCT}$	0.0045	0.0044	0.0043
Case 1	0.0041	0.0040	0.0039
Case 2	0.0038	0.0037	0.0036
Case 3	0.0036	0.0035	0.0034
Case 4	0.0034	0.0033	0.0032
Case 5	0.0033	0.0032	0.0031
Min $JMCC_{BCC}$	0.0031	0.0031	0.0030

**Fig. 15.** Battery capacity for the selected cases versus charging cycles.

charging time for Case 5 increases to 22.10 min (a 41% increase), but the battery average temperature and total charging cost reduce to 26.41 °C (a 4.5% decrease) and 0.0033 Yuan (a 19.5% decrease), respectively. This is mainly due to the fact that a higher average charging current rate favours a faster charging speed. Both the average battery SOC and temperature will also increase dramatically by shortening the charging time. Then the electricity-loss and battery capacity degradation would be also accelerated, further resulting in larger economic charging cost. Besides, there exists a range that linearly increasing the charging time will significantly reduce the charging cost and average battery temperature. Outside this range, further linearly increasing the charging time has a less effect on the decrease of the charging cost and battery temperature. To be specific, as the $JMCC_{BCT}$ increases linearly till Case 3, both $JMCC_{BAT}$ and $JMCC_{BCC}$ dramatically reduce to 26.92 °C

(a 5.1% decrease) and 0.0036Yuan (a 20% decrease). However, after Case 4, further linearly increasing $JMCC_{BCT}$ has less effects on the reductions of $JMCC_{BAT}$ (a 1.8% decrease till Case 5) and $JMCC_{BCC}$ (an 8.3% decrease till Case 5). The MCC profiles before Case 3 can be applied to efficiently reduce the economic cost and battery average temperature during charging.

In order to further analyse the optimization results in terms of $JMCC_{BAT}$ and $JMCC_{BCC}$, battery temperature indicators including the maximum surface temperature $MaxT_s$ and maximum internal temperature $MaxT_l$, along with objective functions $JMCC_{BAC}$ and $JMCC_{BEC}$ corresponding to all specific cases, are illustrated in Table 7. By formulating the temperature constraints in the optimization process, the maximum battery temperatures during charging are all limited within safe boundaries. Interestingly, even though the whole charging time increases gradually from Case 1 to Case 5, the value of $JMCC_{BEC}$ presents an inversely decrease tendency. This is mainly caused by a smaller current rate applied to achieve an increased charging time, and the electricity-loss cost during the total charging process is mainly affected by the charging current rather than the consumed charging time. Therefore, the accumulated $JMCC_{BEC}$ is also decreased.

From the Table 7, it is interesting to note that the values of $JMCC_{BAC}$ in all cases are basically 5 times larger than the $JMCC_{BEC}$, which means that the economic cost caused by battery aging is much larger than that caused by electrical energy loss during one MCC charging. The electricity-loss cost for a cell in one charging cycle is relatively small (here in the condition of on-peak electricity price, the value of $JMCC_{BEC}$ for the minimum charging time case (Min $JMCC_{BCT}$) is 7.670e⁻⁴ Yuan, while the value of $JMCC_{BAC}$ in this case is 0.0038 Yuan). Normally, there are always hundreds or thousands of cells connected in parallel and/or series configuration to support enough power for an EV application. These cells require to be recharged day-by-day. Therefore, the summed electricity-loss cost for an EV is also large enough, and $JMCC_{BEC}$ for each cell cannot be ignored in the study of charging cost. It is noted that $JMCC_{BEC}$ is also affected by the electricity price in different time instants during a day.

Table 8 illustrates the values of the total charging cost $JMCC_{BAC}$ for the selected cases in different electricity-price conditions. It is pronounced that charging the battery in off-peak pattern can reduce the total battery economic cost. The values of $JMCC_{BAC}$ under the on-peak electricity-price condition are usually 5% more than those under the off-peak condition. As the required charging time increases, differences of charging costs of three electricity-price conditions reduce gradually.

Additionally, although $JMCC_{BAC}$ is chosen to analyse the economic charging cost during the same electricity-price range in this study, it is still easy to apply this charging cost function for analysing different electricity-price ranges.

The battery capacity fading levels for all selected cases over charging number are plotted in Fig. 15. Interestingly, as the required charging time for each cycle increases from Case 1 to Case 5, the battery capacity degradation will slow down. At the first 70 charging cycles, the battery capacity degradations for all charging cases are almost similar. After that, the corresponding capacity degradations among these charging cases present obviously different tendencies. Quantitatively, in the case of the minimum charging cost ($MinJMCC_{BCC}$), the battery capacity degrades to 2.478Ah (99.12%) till the first 70 charging cycle, and 2.231Ah (89.21%) after 1200 charging cycle. While the battery capacity degradation for the case of the minimum charging time ($MinJMCC_{BCT}$) is 2.471Ah, just 0.28% less than the former till the first 70 charging cycle. But this value becomes 2.226Ah after 1200 charging cycle, which is nearly 9.92% less than the former. Therefore, for the battery economic charging, different MCC charging profiles would have a great influence on the economic cost caused by battery aging especially after multiple charging cycles. In order to achieve economy-aware charging and also finish the charging task as fast as possible, the MCC profile with the large current rate is preferred for a new battery, owing to the fact that differences of capacity fade for fresh batteries are small or even negligible. With the increase in the cycle number of battery charging (i.e., after 70 cycles), the applied MCC profiles need to be carefully selected for economic charging. The battery aging cost function $JMCC_{BAC}$ and the electricity-loss cost function $JMCC_{BEC}$, which are formulated in this study, can be used as useful performance indicators to guide EV owners to select suitable charging profiles for various economic requirements.

5. Conclusions

Battery charging exerts an enormously important role in the public acceptance, energy management, safety and cost competitiveness of EVs. This paper constructs a novel constrained multi-objective optimization framework to achieve economy-conscious battery charging from the viewpoint of EV users, where three crucial but contradictory objectives including the charging time, battery average temperature, and particularly charging cost are formulated. Then the NSGA-II approach is employed to optimize the multistage constant-current (MCC) charging profile with the purpose of enabling energy and economic management for Li-ion battery charging. Hard constraints on the battery SOC, charging current, terminal voltage, and temperature are all explicitly considered in the optimization process. Furthermore, the effects of the cut-off voltage, convection resistance, and ambient temperature on the charging performance are comparatively studied via Pareto frontier. To our knowledge, this is for the first time to simultaneously consider both the electricity-loss cost and battery aging cost in the context of directly designing battery charging profiles. The proposed charging management strategy has been examined in two extreme cases including the minimum time charging, minimum economic-loss charging, as well as five cases of balanced charging. Some conclusions can be obtained as:

- 1) The objectives of charging cost and battery average temperature both conflict with the battery charging time.
- 2) Small values of cut-off voltage, heat convection resistance and ambient temperature result in low economic charging cost and battery average temperature by sacrificing charging speed, and vice versa.
- 3) Battery economic cost during one charging caused by the capacity degradation is nearly five times larger than that caused by the electricity-loss for the case of Beijing peak-valley TOU price.
- 4) In order to charge battery as fast as possible without incurring large economic charging cost, the MCC profile with the large current rate is preferred for the fresh battery (before 70 cycles).

- 5) For a whole life cycle of the A123 battery through using MCC pattern, total cost of the minimum economic loss charging is 13.2 Yuan, which is almost 21% less than that from the minimum time charging case.
- 6) The proposed framework could also be equally applicable to energy and economic management of other battery types with appropriate battery dynamic models.

Declaration of interest statement

None declared.

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